Yassine Ben Maktouf

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Summary

Data Science Engineer and Actuary with experience in machine learning, deep learning, NLP/LLM systems, and risk modeling. Skilled in Python, SQL, TensorFlow, and PyTorch with strong problem-solving, analytical, and teamwork abilities. Passionate about applying AI and data-driven methods to solve real-world challenges and contribute to innovative projects.

Experience

- **End-of-Studies Internship (PFE) at Vermeg, Tunisia** *Feb 2026 – Jul 2026*
  - Built an AI-assisted copilot that configures PERI retirement-insurance products in Vermeg's Solife platform from specification documents, under an "extract then apply" design: an LLM extracts intent while a deterministic, auditable engine (lxml/XPath, layered safety guards) applies every change to regulated XML and Groovy business rules.
  - Developed a hybrid RAG assistant (all-MiniLM dense retrieval + BM25 with Reciprocal Rank Fusion) over an actuarial/insurance corpus; benchmarked embedding models and retrieval strategies with MRR/Recall/NDCG to select the production configuration.
  - Ran evaluation studies (fine-tuned XLM-RoBERTa vs. LLM for NER, NLI-based faithfulness auditing) and applied grammar-constrained decoding to improve extraction reliability; delivered a React + FastAPI multi-instance platform with JWT auth and per-instance retrieval isolation.
- **Internship at MEDIS, Tunisia** *Jun 2025 – Aug 2025*
  - Developed a sales forecasting system using ML and DL models, selecting and optimizing the best-performing approach.
  - Built an automated data pipeline with SQL Server integration.
  - Delivered a web-based application with interactive reporting and visualization tools to support business decision making.
- **Internship at STIP, Tunisia** *Jun 2024 – Jul 2024*
  - Designed an SQL Server database and REST APIs using Node.js for managing employees, phone numbers, and contracts.
  - Developed a user friendly interface with React.js and Bootstrap for managing assignments.
  - Centralized data management, automatic phone number assignment, and change tracking with role-based access control.

Key Projects

- **Computer Vision and OCR Medical Documents System:** Engineered an AI system for multilingual document processing (French/Arabic) using YOLO-cls for classification, fine-tuned OCR, zero-shot layout detection, and a RAG-based LLM, with web scraping via Selenium. Deployed on FastAPI with Docker and Kafka for scalable performance.
- **Churn Prediction API:** Developed an attrition analysis pipeline using supervised (Random Forest, Decision Tree with SMOTE) and unsupervised methods (PCA, K-Means), deployed with joblib, and exposed through REST APIs.
- **Audio Emotion Recognition:** Created a diffusion model with ResNet-50 to enhance audio emotion, tested on EmoDB and RAVDESS datasets, and developed a Python pipeline for mel-spectrogram generation.
- **Actuarial Portfolio Heterogeneity Study:** Modeled mortality rates for French smokers/non-smokers (1950 cohort) using Lee-Carter in R, projected trends, and designed fair annuity pricing strategies.

Education

- **Le Mans University, France** – Master's degree in Actuarial Science *2024 – 2026*
- **ESPRIT, Tunisia** – Data Science Engineering *2023 – 2026*
- **IPEIT, Tunisia** – Pre-engineering in Mathematics-Physics *2021 – 2023*
- **Lycée Ibnou Rachik** – Baccalaureate in Mathematics *2021*

Skills & Certifications

Skills:
Programming Languages: Python, Java, R, JS, SQL, C, C#
Frameworks/Libraries: SpringBoot, .NET, React, Flask, FastAPI, Tailwind CSS
Machine Learning/AI: Machine Learning, Deep Learning, TensorFlow, Scikit-learn, PyTorch, Keras, YOLO, HuggingFace Transformers, NLP, LLM, RAG
DevOps: Docker, GitHub Actions, MLflow, MLOps, CI/CD
Databases:SQL Server, MongoDB, MySQL, SQLite

Certifications:
IBM Machine Learning with Python
IBM Deep Learning with Keras
IBM Artificial Intelligence Foundations
NVIDIA AI for Anomaly Detection
NVIDIA Large Language Models

Languages:
Arabic, English, French